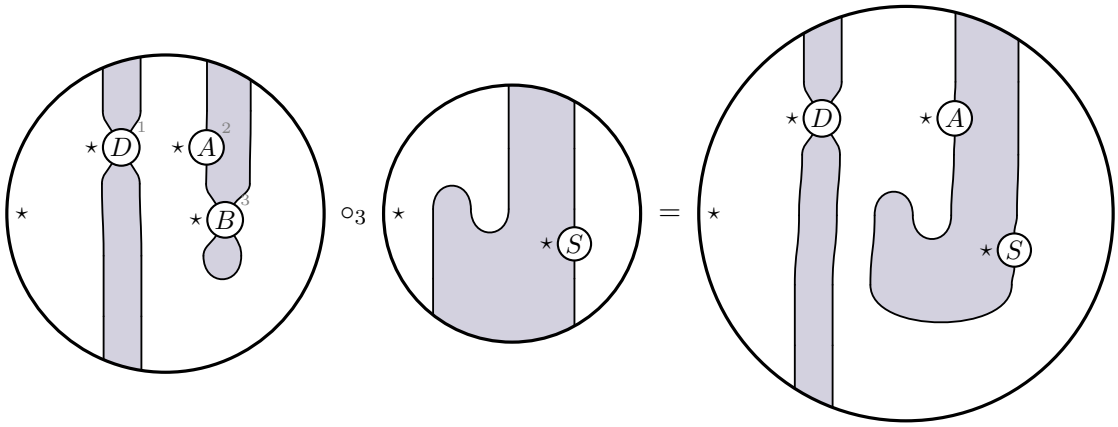
The planaralgebra package

Planar tangles and planar algebra diagrams in LaTeX with Ti*k*Z

Version 1.0, 2026/10/30

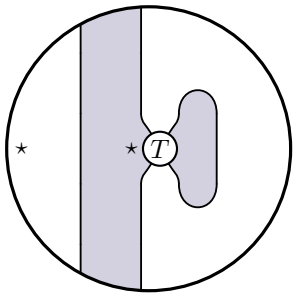
A planar tangle is a disc with some input discs removed, together with disjoint strings that connect the marked points on the boundaries; the regions between the strings are shaded in a checkerboard fashion, and every disc has a star $\star$ that marks its first region. Planar tangles act on planar algebras by composition: $S \circ_i T$ glues the tangle T into the i-th input disc of S. This package draws such tangles from a description layer by layer, in the style of the plumber package for 2d cobordisms:



Each `\PA{`⟨*m*,*n*⟩`}{`⟨*items*⟩`}` is one layer, the side-by-side juxtaposition of the listed strings and input discs, going from m to n boundary points; consecutive `\PA` commands are stacked into one tangle. The package places the strings, shades the regions, draws the stars and the outer disc, and with `reduce` it computes a normal form of the tangle: closed loops are removed and counted, and the strings are pulled tight.

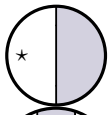
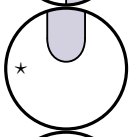
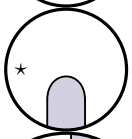
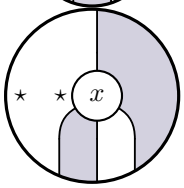
1 Quick start

```
\usepackage{planaralgebra}   % loads TikZ
...
\PA{2,4}{id, id, coev}\PA{4,4}{id, T(2,2), id}\PA{4,2}{id, id, ev}
```



The first argument is the number of boundary points above and below the layer (`2,4` means two strands in, four out); it is checked against the items and may also be left empty. The second argument is a comma-separated list of items, read from left to right (from top to bottom in horizontal diagrams). `id^3` is short for `id, id, id`, and the same works for any item.

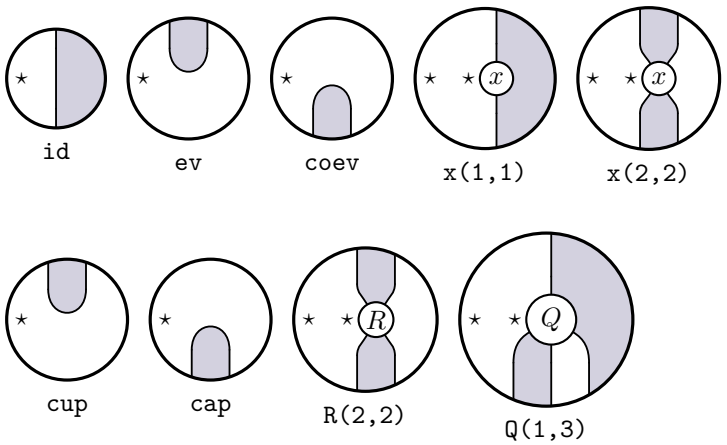
2 Generators and input discs

item	aliases	strands	picture
`id`	`line`, `strand`	$1 \to 1$	
`ev`	`cup`, `pair`	$2 \to 0$	
`coev`	`cap`, `copair`	$0 \to 2$	
⟨*label*⟩`(a,b)`		$a \to b$	

Every item that is not a generator is an input disc. Its label is typeset in math mode, so `\alpha(1,1)` gives a disc labelled α. The counts `(a,b)` say how many strands leave the disc on the input side (above it) and on the output side (below it); `(a;b)` and `(a/b)` are accepted as well. When a layer has a single disc and its counts are declared, the counts of the disc may be omitted: `\PA{2,2}{x}` is the same as `\PA{2,2}{x(2,2)}`. Discs can be repeated like generators: `f(1,1)^2`.

`\PAgenerators[`⟨*options*⟩`]` draws the items of the key `list` (by default `id`, `ev`, `coev`, `x(1,1)` and `x(2,2)`) next to each other; the starred form leaves out `id`. With `names` each one gets its name.

```
\PAgenerators[names]
\PAgenerators*[names, list={cup, cap, R(2,2), Q(1,3)}]
```

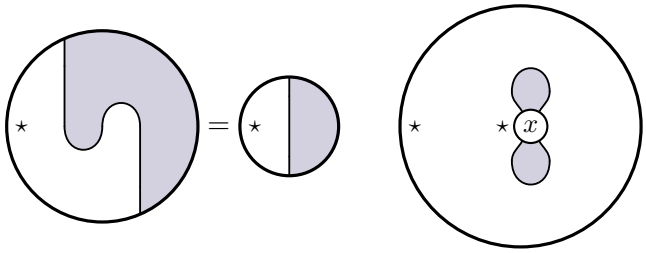


3 The features

3.1 Composition

Consecutive `\PA` commands form one tangle (spaces and single line breaks between them are allowed; an empty line or any other token ends the tangle). The output points of one layer are glued to the input points of the next one, so the counts must agree. Options can be given to any `\PA` of the chain and apply to the whole tangle.

```
\[ \PA{1,3}{id, coev}\PA{3,1}{ev, id} = \PA{1,1}{id}
   \qquad
   \PA{0,2}{coev}\PA{2,2}{x}\PA{2,0}{ev} \]
```

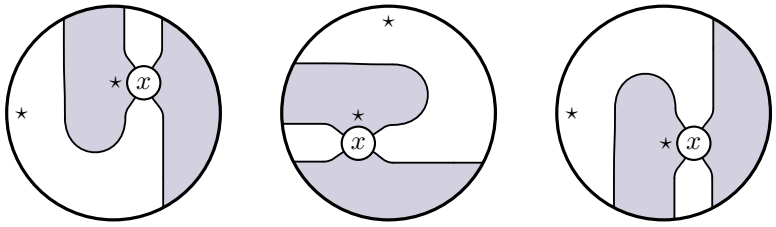


If the counts do not match, or a disc has no counts that could be inferred, the package reports an error that names the offending layer and draws a red ?? instead of the tangle. Should `\PA` already be defined by another package, `\PAlayer` is a synonym with the same arguments (the two can be mixed in a chain).

3.2 Vertical or horizontal

Tangles are `vertical` by default: composition runs from top to bottom and the first strand is on the left. With `horizontal`, composition runs from left to right. `reverse` flips the direction of composition (bottom to top, or right to left). `orientation=vertical` and `orientation=horizontal` are long forms, and `\paset{`⟨*options*⟩`}` changes the defaults for the rest of the group.

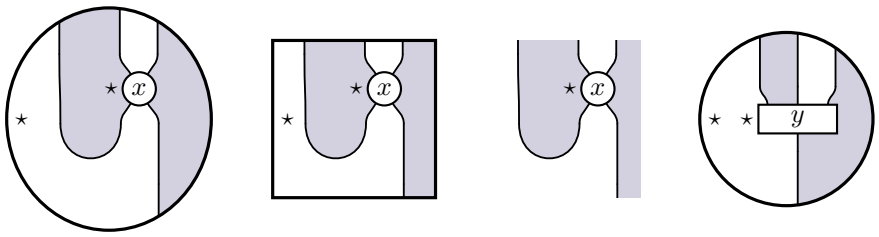
```
\PA{3,3}{id, x(2,2)}\PA{3,1}{ev, id}
\PA[horizontal]{3,3}{id, x(2,2)}\PA{3,1}{ev, id}
\PA[reverse]{3,3}{id, x(2,2)}\PA{3,1}{ev, id}
```



3.3 The outer disc

The key `outer` chooses the shape of the boundary: `circle` (the default), `ellipse`, `rectangle` or `none`. `box shape` (`circle` or `rectangle`) does the same for the input discs; wide labels make rectangular discs longer.

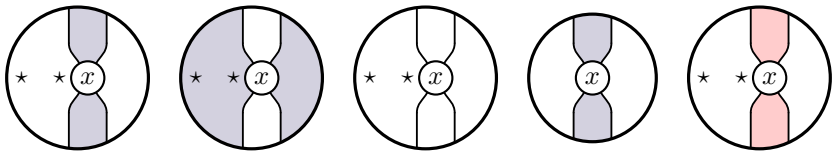
```
\PA[outer=ellipse]{3,3}{id, x(2,2)}\PA{3,1}{ev, id}
\PA[outer=rectangle]{3,3}{id, x(2,2)}\PA{3,1}{ev, id}
\PA[outer=none]{3,3}{id, x(2,2)}\PA{3,1}{ev, id}
\PA[box shape=rectangle]{3,1}{y(3,1)}
```



3.4 Shading and stars

The regions of a tangle are shaded in a checkerboard fashion. With `sign=+` (or `positive`, the default) the region at the star of the outer disc, the leftmost region, is unshaded; `sign=-` (or `negative`) shades it. `shading=false` turns the shading off; if a disc has an odd number of strands, no checkerboard shading exists and the package warns and draws the tangle without it. Every disc gets a star in its first region, and `stars=false` leaves the stars out. `shade` sets the colour of the shaded regions.

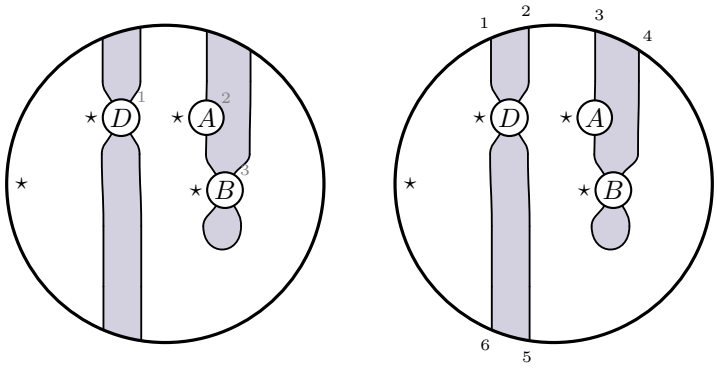
```
\PA{2,2}{x}  \PA[negative]{2,2}{x}  \PA[shading=false]{2,2}{x}
\PA[stars=false]{2,2}{x}  \PA[shade=red!20]{2,2}{x}
```



3.5 Numbers and labels

`numbers` writes the number of every input disc into it (instead of its label), in the order in which the discs appear in the layers. These are the numbers used by `insert` below. `labels` numbers the boundary points of the outer disc, clockwise from the star.

```
\PA[numbers]{4,4}{D(2,2), A(1,1), id}\PA{4,4}{id, id, B(2,2)}\PA{4,2}{id, id, ev}
\PA[labels]{4,4}{D(2,2), A(1,1), id}\PA{4,4}{id, id, B(2,2)}\PA{4,2}{id, id, ev}
```

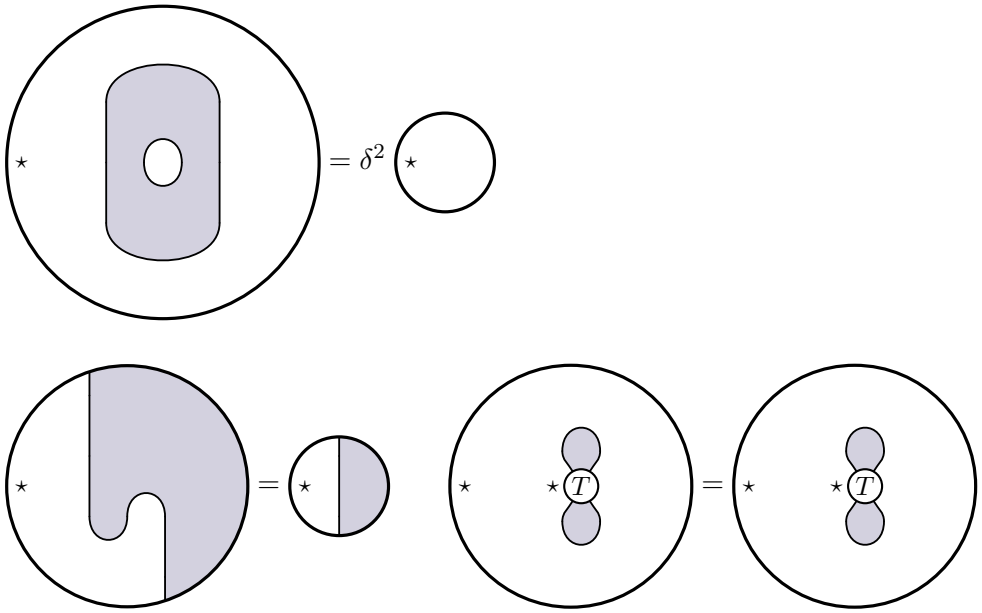


3.6 Reduced form

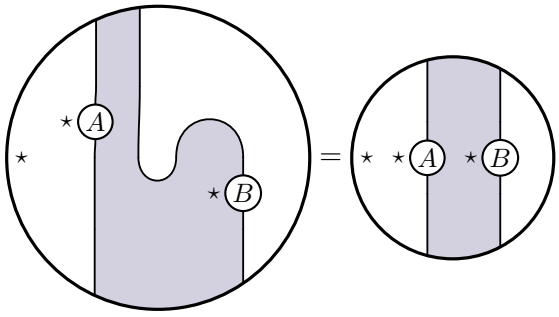
Two tangles describe the same element of a planar algebra when they are isotopic, and a closed loop can be replaced by the scalar δ, the loop value. With `reduce` the package computes a normal form: it removes closed loops (they become a coefficient δ^k in front of the tangle), pulls the strings tight so that zig-zags disappear and strands that only turn around next to a disc are retracted, and draws the result in a standard shape. In the normal form the input discs are stacked one above the other, and

between two consecutive discs (and above the first and below the last) the strings form a Temperley–Lieb diagram written as cups, then through strands, then caps. Discs on neighbouring levels can often be interchanged, one passing the other on the left or on the right; the package goes through all the arrangements of the discs that can be reached in this way and takes the one whose description comes first in a fixed (lexicographic) order. So isotopic tangles give the same normal form, and reducing a normal form gives it back. Tangles with many discs that can move independently of each other (for instance several discs that are not connected to the boundary) have many arrangements; the search stops after `max states` of them (default `100`) with a warning, and then the normal form may depend on how the tangle was entered.

```
\[ \PA{0,2}{coev}\PA{2,4}{id, coev, id}\PA{4,2}{id, ev, id}\PA{2,0}{ev}
   = \PA[reduce]{0,2}{coev}\PA{2,4}{id, coev, id}\PA{4,2}{id, ev, id}\PA{2,0}{ev} \]
\[ \PA{1,1}{id}\PA{1,3}{id, coev}\PA{3,1}{ev, id}
   = \PA[reduce]{1,1}{id}\PA{1,3}{id, coev}\PA{3,1}{ev, id}
   \qquad
   \PA{0,2}{coev}\PA{2,2}{T(2,2)}\PA{2,0}{ev}
   = \PA[reduce]{0,2}{coev}\PA{2,2}{T(2,2)}\PA{2,0}{ev} \]
```

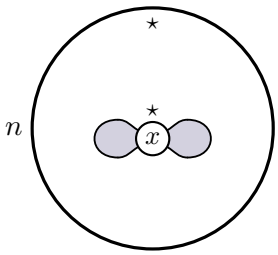



```
\[ \PA{2,4}{A(1,1), id, coev}\PA{4,2}{id, ev, B(1,1)}
   = \PA[reduce]{2,4}{A(1,1), id, coev}\PA{4,2}{id, ev, B(1,1)} \]
```



The key `loop value` sets the symbol of a loop (default `\delta`), and `coefficient=false` draws the reduced tangle without the coefficient.

```
\PA[horizontal, reduce, loop value=n]{0,2}{coev}\PA{2,0}{ev}\PA{0,2}{coev}\PA{2,2}{x}\PA{2,0}{ev}
```



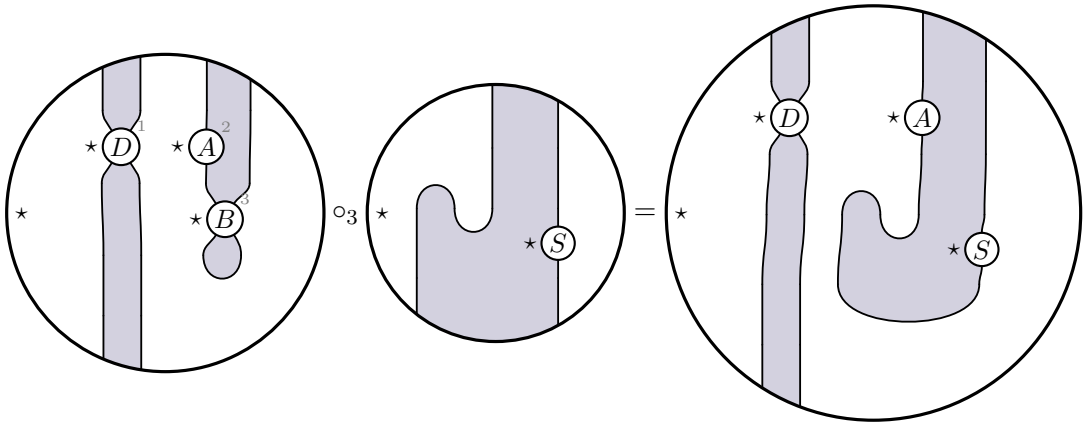
After every tangle (reduced or not) the following macros expand to data of the tangle that was just drawn: `\PAlastloops` (the number of closed loops), `\PAlastboxes` (the number of input discs) and, after `reduce`, `\PAlastnormalform` (the normal form as a string of `\PA` commands, which can be fed back to the package). For the last example, `\PAlastloops` is 1, `\PAlastboxes` is 1, and the normal form is

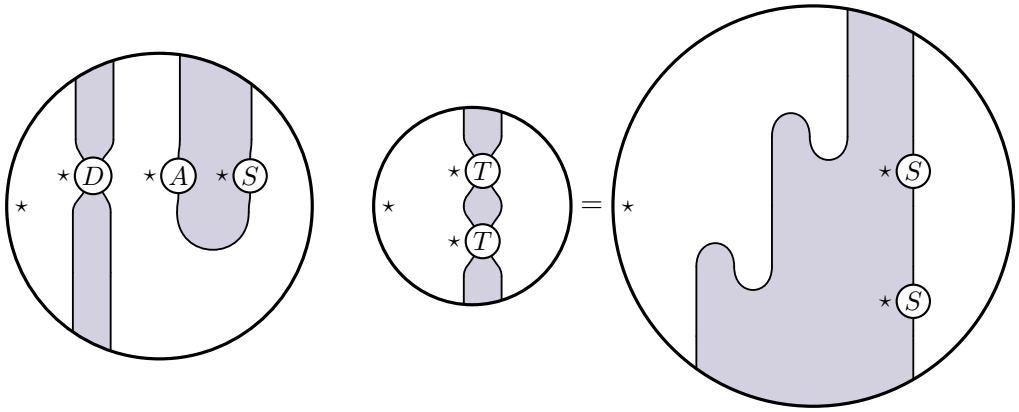
```
\PA{0,2}{coev}\PA{2,2}{x(2,2)}\PA{2,0}{ev}
```

3.7 Defining and composing tangles

`\PAdefine[`⟨*options*⟩`]{`⟨*name*⟩`}{`⟨*layers*⟩`}` stores a tangle under a name instead of drawing it (with `reduce`, its normal form is stored). The key `insert={`*i*`=`⟨*name*⟩`, ...}` glues the named tangle into the input disc number *i* (as shown by `numbers`); its numbers of strands must agree with those of the disc. With `expand`, every input disc whose label is the name of a tangle is replaced by it.

```
\PAdefine{T}{\PA{2,4}{coev, id, id}\PA{4,2}{id, ev, S(1,1)}}
\[ \PA[numbers]{4,4}{D(2,2), A(1,1), id}\PA{4,4}{id, id, B(2,2)}\PA{4,2}{id, id, ev}
   \circ_3 \PA{2,4}{coev, id, id}\PA{4,2}{id, ev, S(1,1)}
   = \PA[insert={3=T}]{4,4}{D(2,2), A(1,1), id}\PA{4,4}{id, id, B(2,2)}\PA{4,2}{id, id, ev} \]
\[ \PA[insert={3=T}, reduce]{4,4}{D(2,2), A(1,1), id}\PA{4,4}{id, id, B(2,2)}\PA{4,2}{id, id, ev}
   \qquad
   \PA{2,2}{T}\PA{2,2}{T} = \PA[expand]{2,2}{T}\PA{2,2}{T} \]
```



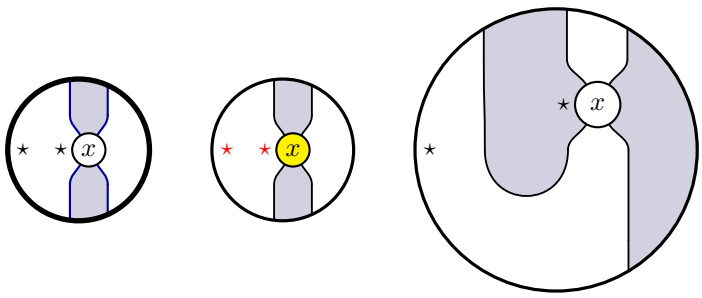


4 Appearance

key	default	meaning
`strand sep`	`0.5`	distance between neighbouring strands (cm)
`length`	`0.8`	length of a layer (cm)
`box size`	`0.22`	minimal radius of an input disc (cm)
`margin`	`0.3`	space between the strings and the outer disc (cm)
`label font`	`\small`	font of the disc labels
`scale`	`1`	scales the picture (not the text)
`baseline`	`-0.5ex`	vertical alignment, as in TikZ
`tikz`		further options for the `tikzpicture`
`shade`		colour of the shaded regions
`separator`		symbol between the pictures of `\PAgenerators`
`gap`	`0.3`	space between the pictures of `\PAgenerators` (cm)

The drawing uses the TikZ styles `pa string`, `pa shaded`, `pa box` (input discs), `pa outer`, `pa star`, `pa label` (disc labels), `pa number` (disc numbers), `pa boundary label`, `pa name`, `pa separator`, `pa coefficient` and `pa error`. The keys `string style`, `shade style`, `box style`, `outer style` and `star style` append to the corresponding styles.

```
\PA[string style={thick, blue!60!black}, outer style={line width=2pt}]{2,2}{x}
\PA[box style={fill=yellow}, star style={text=red}]{2,2}{x}
\PA[length=1.2, strand sep=0.8, box size=0.3]{3,3}{id, x(2,2)}\PA{3,1}{ev, id}
```



5 Layout

The numbers of strands change from layer to layer, so their positions are computed. Every item says where it would like the strands next to it to be: a strand keeps its position, a cup or cap puts its two

ends half a strand distance on either side of its middle, and an input disc spreads its strands evenly around its centre. On every seam between two layers the positions are the best least-squares fit to these wishes under the condition that neighbouring strands are at least `strand sep` apart (an isotonic regression), and a few sweeps back and forth through the tangle let both neighbours of a seam have their say. The outer disc is then fitted around the picture, and the strands that reach it are extended to its boundary.

6 The standalone class

With `\documentclass[tikz]{standalone}` every TikZ picture becomes a page of its own. Tangles inside `\[...\]` and `\(...\)` are handled like in the temperleylieb and plumber packages: the whole formula becomes one cropped page. Tangles outside formulas are pages of their own. Other math environments (`$...$`, `equation`) cannot be turned into pages; the package warns and draws the tangle inline, so use `\[...\]` there, or the class option `varwidth`.

7 Summary of commands

`\PA[`⟨*options*⟩`]{`⟨*m,n*⟩`}{`⟨*items*⟩`}` One layer; a chain of layers is one tangle. `\PAlayer` is a synonym.

`\PAgenerators*[`⟨*options*⟩`]` The items in `list` side by side.

`\PAdefine[`⟨*options*⟩`]{`⟨*name*⟩`}{`⟨*layers*⟩`}` Stores a tangle for `insert` and `expand`.

`\paset{`⟨*options*⟩`}` Sets options for the current group.

`\PAlastloops`, `\PAlastboxes` Number of closed loops and of input discs of the last tangle.

`\PAlastnormalform` Normal form of the last reduced tangle.

The package is written in expl3 and needs a LaTeX kernel from 2020-10 or newer and TikZ. It is distributed under the LaTeX Project Public License 1.3c.